# TsaoSciResearcher Engineering Audit

Release 0.7.4 | preflight evidence | 2026-11-02

## Evidence-first scientific research control layer

Question -> evidence -> design -> execution handoff -> validation -> artifact

Truth boundary: a contract or handoff is not an execution receipt.

**341**

Capability contracts

**322**

Named catalog contracts

**15**

Workflows

**20**

Schemas

**67**

Test modules

**7**

Domain packs

## Delivered integration

1. Measurement Boundary with calibration, uncertainty and applicability fields
2. Structure-Property planning with confounders, alternatives and scale bridges
3. Bilingual causality guard and evidence-to-source execution-receipt checks
4. Deterministic test/quality HTML-SVG interfaces and this four-page PDF

### Repository operation

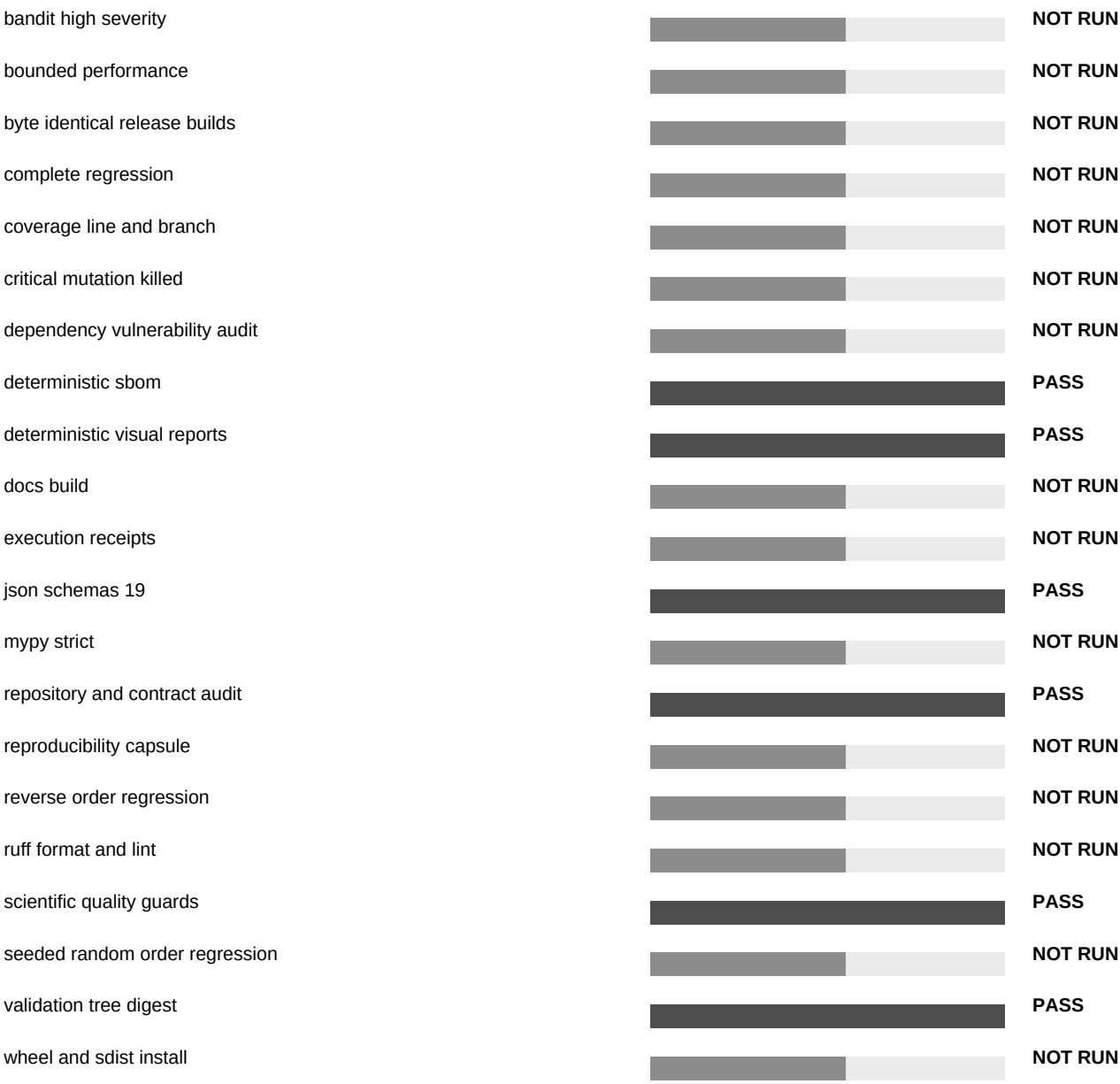
All changes were written directly to main; no branch and no pull request were created.

# Scoped software validation matrix

## Recorded cross-platform baseline

These platform results are a recorded baseline, not a fresh current-tree run.

## Scoped validation gates




**Scoped status applies only to the evidence levels explicitly recorded above.**

# Scientific quality controls

Four deterministic contracts: measurement, multiscale reasoning, causal claims and evidence provenance.

**Examples: PASS 2 | WARN 1 | BLOCK 1 | average completeness 89%**

## Measurement boundary PASS

Declared completeness: 100%

Operational definition, calibration, uncertainty and applicability are explicit.

MB-COMPLETE: Operational measurement boundary is explicit.

Kind: measurement-boundary

## Structure-property planner PASS

Declared completeness: 100%

The chain records intervention, structure, mediator, response and competing...

SP-CHAIN: Structure, mediator and property are all explicit.

Kind: structure-property-plan

## Causality guard BLOCK

Declared completeness: 57%

Association-only evidence blocks causal wording in English and Chinese.

CG-OVERCLAIM: Causal wording exceeds the declared study design.

Kind: causality-guard

## Evidence traceability WARN

Declared completeness: 100%

Claims link to evidence locators and execution claims require receipts.

ET-UNVERIFIED: Locators and receipt IDs are declarations until checked against a real...

Kind: evidence-traceability

BLOCK is intentional when wording exceeds evidence or an execution claim lacks a receipt.

The guards remain configurable and do not hard-code material-specific trends.

# Scoped evidence and limitations

## Recorded validation scope

Scope: preflight

Only the explicitly recorded checks may be interpreted as passing.

## Current-tree end-to-end CI

Status: not recorded

No exact commit attestation is asserted until external CI artifacts record it.

## Explicit limitations

1. Checked-in preflight evidence is not a substitute for the external CI attestation.

2. External scientific execution requires a checksum-verifiable execution receipt.

## Reproduction entry points

python scripts/build_validation_evidence.py --check

python scripts/build_test_dashboard.py --check

python scripts/build_research_quality_dashboard.py --check

python scripts/build_engineering_report.py --check

**Software validation is not acceptance of experiments, external calculations, legal or safety claims.**